

David Chen, Ph.D.

Principal AI & Machine Learning Systems Engineer

[✉ david.chen@example.com](mailto:david.chen@example.com) [☎ +1 \(555\) 456-7890](tel:+15554567890) [📍 New York, NY \(Remote\)](https://www.linkedin.com/in/david-chen-ai) [🔗 linkedin.com/in/david-chen-ai](https://www.linkedin.com/in/david-chen-ai)

[📁 github.com/dchen-ml](https://github.com/dchen-ml)

PROFESSIONAL SUMMARY

Distinguished ML systems engineer with 9+ years of research and production experience in real-time speech processing, high-scale LLM evaluation pipelines, and vector semantic retrieval. Expert in Whisper STT, Kokoro TTS, pgvector HNSW indexing, and anti-jailbreak prompt rails.

CORE TECHNICAL COMPETENCIES

AI & ML Frameworks: PyTorch • Transformers • ONNX Runtime • TensorRT • CUDA • Python

Audio & Speech: Whisper STT • Kokoro TTS • WebRTC Media • Voice Activity Detection (VAD)

Vector & LLM: pgvector (HNSW) • FastEmbed • Structured JSON Output • Prompt Injection Rails

Infra & Systems: Docker ML Images • GPU Resource Pooling • Go/Python Interop • FastAPI

PROFESSIONAL EXPERIENCE

Principal AI Engineer — Synthetix AI Systems

2019 – Present | New York, NY

- Designed sub-300ms real-time voice-to-voice interview streaming pipelines integrating Whisper STT and Kokoro TTS.
- Architected semantic vector search infrastructure in PostgreSQL using pgvector HNSW indexing across 2M+ candidate profiles.
- Engineered deterministic prompt injection defenses and structured JSON evaluation rails achieving 99.8% schema adherence.
- Authored CUDA kernel and ONNX optimizations reducing model inference latency by 3.5x on standard GPU instances.

Senior Machine Learning Researcher — DeepAcoustics Labs

2016 – 2019 | Palo Alto, CA

- Published 4 peer-reviewed papers on acoustic feature representation and neural speech enhancement.
- Built distributed PyTorch training pipelines scaling acoustic models across 64-GPU clusters.

EDUCATION

Ph.D. in Computer Science (AI & NLP) — Stanford University

2017

B.S. in Computer Engineering — Tsinghua University

2013